

Crime Records of India

Database Schema Documentation

Enterprise Database Architecture & Table Documentation

1. Database Overview

The Crime Records of India platform uses MySQL as the primary relational database. The architecture is divided into Authentication Tables, Governance Tables, Analytics Tables, and Dynamic Dataset Tables.

2. Database Architecture Layers

Layer	Purpose
Authentication Layer	Stores users, admins, refresh tokens
Analytics Layer	Tracks searches, views, activity logs
Governance Layer	Tracks admin modifications and deletions
Dataset Layer	Stores crime datasets dynamically

3. Authentication Tables

3.1 users Table

Column	Type	Description
id	INT	Primary key
username	VARCHAR	Unique username
email	VARCHAR	User email
password_hash	TEXT	Encrypted password
full_name	VARCHAR	User full name
last_login	DATETIME	Last login timestamp
created_at	TIMESTAMP	Account creation time

3.2 admins Table

Column	Type	Description
id	INT	Primary key
username	VARCHAR	Admin username
email	VARCHAR	Admin email
password_hash	TEXT	Encrypted password
role	VARCHAR	Admin role
status	VARCHAR	Account status
last_login	DATETIME	Last login timestamp
created_at	TIMESTAMP	Creation time

3.3 refresh_tokens Table

Column	Type	Description
id	INT	Primary key
user_type	VARCHAR	user/admin

user_id	INT	Owner ID
token	TEXT	JWT refresh token
expires_at	DATETIME	Expiration timestamp
created_at	TIMESTAMP	Creation time

4. Analytics Tables

4.1 *views_log* Table

Column	Purpose
user_id	Tracks viewer
username	Viewer username
table_viewed	Dataset viewed
user_ip	Viewer IP
created_at	View timestamp

4.2 *search_logs* Table

Column	Purpose
user_id	Tracks searching user
table_name	Dataset searched
search_filters	Applied filters
result_count	Returned records
user_ip	Requester IP
created_at	Search timestamp

5. Governance Tables

Governance tables track administrative actions for transparency, accountability, and auditability.

5.1 *admin_modification_logs*

Column	Purpose
admin_id	Admin identifier
admin_username	Admin username
table_name	Affected dataset table
action_type	UPDATE / DELETE
old_data	Previous dataset values
new_data	Updated values
changed_columns	Modified columns
ip_address	Admin IP address
created_at	Action timestamp

6. Dynamic Dataset Tables

The platform dynamically exposes crime datasets through metadata-driven APIs. Each dataset table can be queried using dynamic routes.

Dataset Table
auto_theft
victims_of_rape
serious_fraud
police_housing
property_stolen_and_recovered
human_rights_violation_by_police
specific_purpose_of_kidnapping_and_abduction
trial_of_violent_crimes_by_courts

7. Redis Cache Layer

Redis is used as an in-memory cache layer to improve performance for metadata APIs and analytics endpoints.

Cache Key	Purpose	TTL
all_tables	Caches dataset table list	30 Minutes
schema_{table}	Caches dataset schema	1 Hour
analytics_overview	Caches analytics dashboard	5 Minutes

8. Database Relationships

- users.id → refresh_tokens.user_id
- admins.id → refresh_tokens.user_id
- users.id → views_log.user_id
- users.id → search_logs.user_id
- admins.id → admin_modification_logs.admin_id